


Aggregate Production Function

$$\text{Aggregate output} = \underbrace{\text{GDP}}_{\text{"real" terms}} = Y = \text{Real Income}$$

K = Capital Stock I
= sum of all Investment over time
in Equipment, Structures, and
Software net of depreciation

N = # people working # hours each person works
extensive margin for N-input intensive margin for N-input

Z = Total Factor Productivity
TFP = all stuff that
makes GDP that isn't
 K or N

$$Y = Z K^{\alpha} N^{\beta} \quad \left. \vphantom{Y = Z K^{\alpha} N^{\beta}} \right\} \begin{array}{l} \text{Cobb-Douglas} \\ \text{Production} \\ \text{Function} \end{array}$$

$$Y = \underbrace{wN}_{\substack{\text{Aggregate} \\ \text{wage bill} \\ \text{where } w = \text{real} \\ \text{wage}}} + \underbrace{rK}_{\substack{\text{capitalist's} \\ \text{real income}}}$$

$$wN + rK = Z K^{\alpha} N^{\beta}$$

$$\frac{d[wN + rK]}{dN} = \frac{d[Z K^{\alpha} N^{\beta}]}{dN}$$

$$w + d = \beta Z K^{\alpha} N^{\beta-1} = \text{marginal productivity of } N$$

$$w = \beta \frac{Y}{N} \quad \text{solve for } \beta$$

$$\frac{wN}{Y} = \text{N-share of } Y = \beta \quad \left. \vphantom{\frac{wN}{Y} = \beta} \right\} \begin{array}{l} \text{estimated} \\ \text{in data} \end{array}$$

$$.7 \approx \beta$$

By symmetry,

$$r = \alpha \frac{Y}{K}$$

$$\frac{rK}{Y} = \alpha = .3$$

$$\alpha + \beta = 1$$

$$Y = Z K^{.3} N^{.7}$$

variant of
Cobb
Douglas

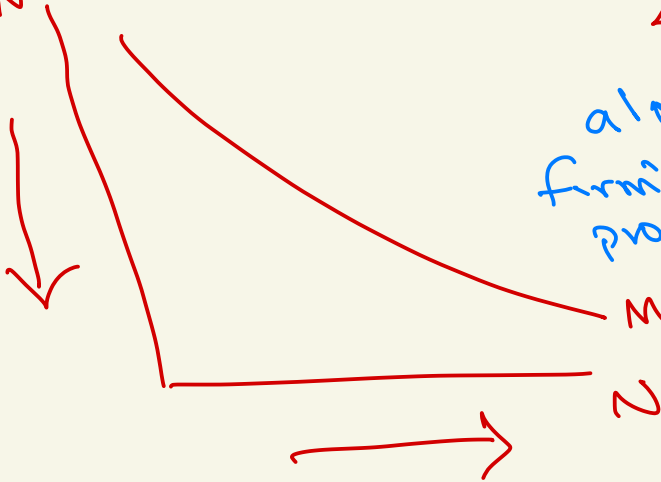
Learn: Diminishing returns to N

$$\frac{dY}{dN} = .7 Z K^{.3} N^{.7-1}$$

$$\underbrace{\text{MPN}} = .7 Z K^{.3} N^{-.3} > 0$$

$$\begin{aligned} \frac{d\text{MPN}}{dN} &= \frac{d[.7 Z K^{.3} N^{-.3}]}{dN} \\ &= \underbrace{-.3 (.7) Z K^{.3} N^{-1.3}}_{< 0} \end{aligned}$$

$\text{MPN} = w$



along this curve,
firm's marginal
profits are zero

$\text{MPN} = Nd$

Learn: Cobb-Douglas Production
Functions exhibit Constant Returns
to Scale (if scale up both
K and N by same
amount)

$$Y_1 = Z K^{.3} N^{.7} \quad .3 + .7 = 1$$

$$\begin{aligned} Y_5 &= Z (5K)^{.3} (5N)^{.7} \\ &= \underbrace{5^{.3} 5^{.7}}_{5^{.3+.7}=5} \underbrace{Z K^{.3} N^{.7}}_{Y_1} \end{aligned}$$

$$Y_5 = 5 \cdot Y_1$$

One more nice thing about
Cobb-Douglas:

$$\frac{GDP}{N} \quad \left\{ \begin{array}{l} \text{roughly per capita} \\ \text{GDP (GDP per} \\ \text{population)} \end{array} \right.$$

$$\text{roughly } \frac{GDP}{N} =$$

standard of living

$$\frac{Y}{N} = Z K^{.3} N^{.7} N^{-1}$$

$$= Z K^{.3} \bar{N}^{-.3}$$

$$\frac{Y}{N} = Z \left(\frac{K}{N} \right)^{.3} \quad \left\{ \begin{array}{l} \text{standards} \\ \text{of living} \\ \text{change with} \end{array} \right.$$

technological
progress and
per-capita K growth

In econ $\log = \ln$

$$GDP = Z K^{.3} N^{.7}$$

$$\ln GDP = \ln Z + \ln K^{.3} + \ln N^{.7}$$

* log of product is sum of the logs

$$\ln GDP = \ln Z + .3 \ln K + .7 \ln N$$

$$* \log \text{ of } x^\alpha = \alpha \ln x$$

$$* \frac{d \ln x}{dx} = \frac{1}{x}$$